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G H Raisoni International Skill Tech University, Pune

S.Y. B.Tech
(TERM-III)
CAE I- WINTER 2025 (2024 Pattern)
Computer Architecture and Organization (25BTPCL303)

(Time: - 01 Hour) (Max. Marks: 20)

Instructions to the students:

i) All questions are compulsory. ii) Neat diagrams must be drawn wherever necessary.
iii) Figures to the right indicate full marks. iv) Use of scientific calculator is allowed
v) Assume suitable data, if necessary.

Q.No	Question	Marks	CO	BL
Q1.(a)	Illustrate the main components (Input Unit, CPU, Memory Unit, and Output Unit) and explain the flow of data and control among them.	[5]	CO1	BL2
b)	Describe the fetch, decode, and execute steps of instruction cycle. Draw the diagram showing how CPU processes each instruction for ADD A	[5]	CO1	BL2
	OR			
Q 2.(a)	Differentiate between CISC and RISC architectures with examples	[5]	CO1	BL2
(b)	Identify addressing Modes of Solving instructions & explain same with proper example.	[5]	CO1	BL3
Q 3.(a)	Illustrate Fixed-Point Arithmetic concept with suitable examples.	[5]	CO2	BL2
(b)	Describe floating-point representation, addition, and multiplication operations. With suitable example.	[5]	CO2	BL3
	OR			
Q. 4 (a)	Describe the steps of Booth's multiplication algorithm and solve a multiplication using Booth's method.	[5]	CO2	BL3
(b)	Illustrate and explain how carry look ahead adder improves addition speed over ripple carry adders.	[5]	CO2	BL3